

EXP.N0:2

IMPLEMENTATION OF PLAYFAIR CIPHER

AIM:

To write a C program to implement the Playfair Substitution technique.

ALGORITHM:

STEP-1: Read the plain text from the user.

STEP-2: Read the keyword from the user.

STEP-3: Arrange the keyword without duplicates in a 5*5 matrix in the row order and fill the remaining cells with missed out letters in alphabetical order. Note that „i“ and „j“ takes the same cell.

STEP-4: Group the plain text in pairs and match the corresponding corner letters by forming a rectangular grid.

STEP-5: Display the obtained cipher text.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
#define SIZE 5
```

```
char keyMatrix[SIZE][SIZE];
```

```
void generateKeyMatrix(char key[])
```

```
{
```

```
    int used[26] = {0};
```

```
int i, j = 0, k = 0;
```

```
used['J' - 'A'] = 1;
```

```
for (i = 0; key[i] != '\0'; i++)
```

```
{
```

```
    char ch = toupper(key[i]);
```

```
    if (ch == 'J')
```

```
        ch = 'I';
```

```
    if (!used[ch - 'A'])
```

```
    {
```

```
        keyMatrix[j][k++] = ch;
```

```
        used[ch - 'A'] = 1;
```

```
    if (k == SIZE)
```

```
    {
```

```
        k = 0;
```

```
        j++;
```

```
    }
```

```
}
```

```
}
```

```
for (i = 0; i < 26; i++)
```

```
{  
    if (!used[i])  
    {  
        keyMatrix[j][k++] = i + 'A';  
  
        if (k == SIZE)  
        {  
            k = 0;  
            j++;  
        }  
    }  
}
```

```
void findPosition(char ch, int *row, int *col)
```

```
{  
    int i, j;  
  
    if (ch == 'J')  
        ch = 'I';  
  
    for (i = 0; i < SIZE; i++)  
    {  
        for (j = 0; j < SIZE; j++)
```

```

    {
        if (keyMatrix[i][j] == ch)
        {
            *row = i;
            *col = j;
            return;
        }
    }
}

```

```

void encrypt(char a, char b)

```

```

{
    int r1, c1, r2, c2;

    findPosition(a, &r1, &c1);
    findPosition(b, &r2, &c2);

    if (r1 == r2)
    {
        printf("%c%c", keyMatrix[r1][(c1 + 1) % 5], keyMatrix[r2][(c2 + 1) % 5]);
    }
    else if (c1 == c2)
    {

```

```
        printf("%c%c", keyMatrix[(r1 + 1) % 5][c1], keyMatrix[(r2 + 1) % 5][c2]);
    }

    else

    {

        printf("%c%c", keyMatrix[r1][c2], keyMatrix[r2][c1]);

    }

}
```

```
int main()

{

    char key[100], text[100];

    int i;

    printf("Enter Key: ");

    scanf("%s", key);

    printf("Enter Plain Text: ");

    scanf("%s", text);

    generateKeyMatrix(key);

    printf("\nKey Matrix:\n");

    for (i = 0; i < 5; i++)

    {
```

```

    int j;

    for (j = 0; j < 5; j++)

        printf("%c ", keyMatrix[i][j]);

    printf("\n");
}

printf("\nCipher Text: ");

for (i = 0; text[i] != '\0'; i += 2)
{
    char a = toupper(text[i]);

    char b = (text[i + 1] != '\0') ? toupper(text[i + 1]) : 'X';

    if (a == 'J') a = 'I';

    if (b == 'J') b = 'I';

    if (a == b)

        encrypt(a, 'X');

    else

        encrypt(a, b);
}

return 0;
}

```

OUTPUT:

```
Enter Key: BEST
Enter Plain Text: KESHAVARAM

Key Matrix:
B E S T A
C D F G H
I K L M N
O P Q R U
V W X Y Z

Cipher Text: PDAFBZTUTN

...Program finished with exit code 0
Press ENTER to exit console.█
```

RESULT:

Thus the Playfair cipher substitution technique had been implemented successfully.